

Fundamentals of Business Intelligence



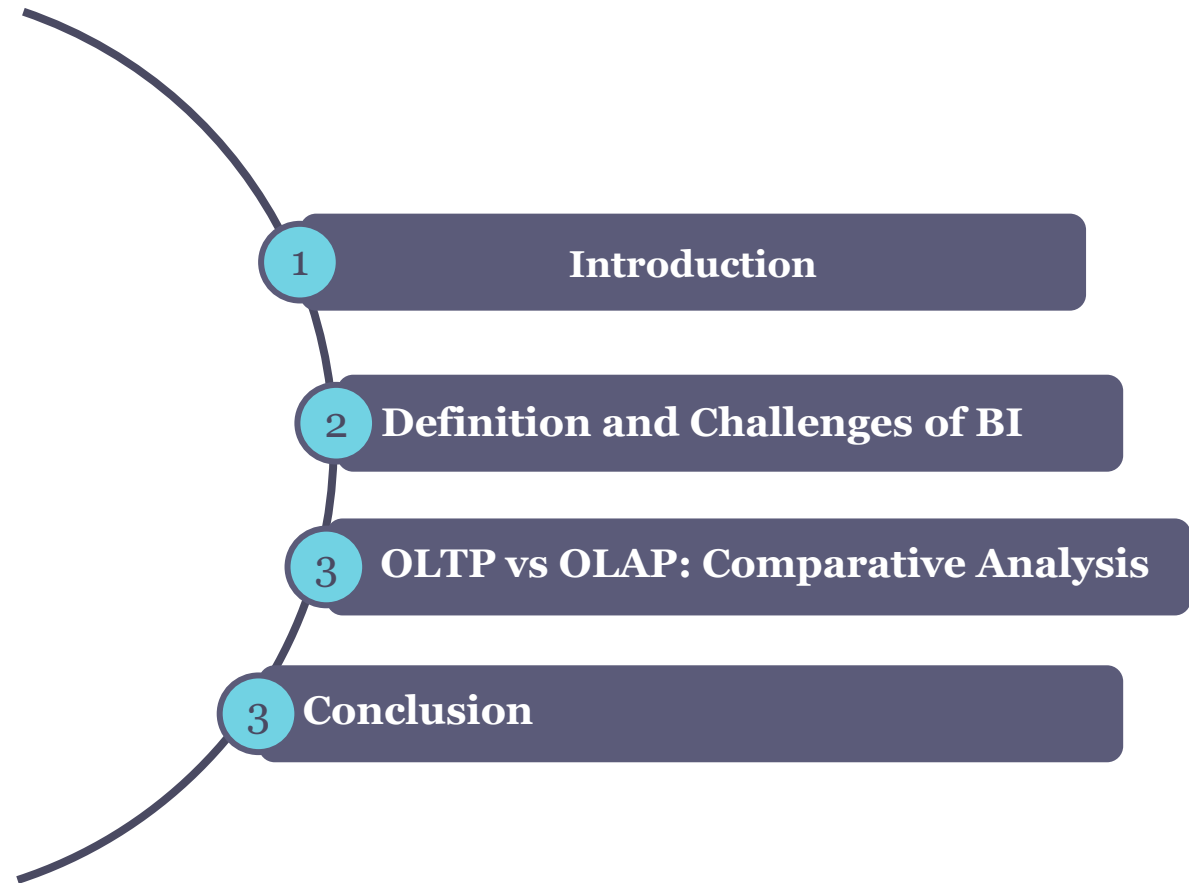


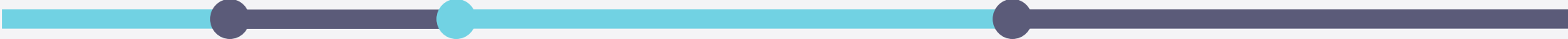
Course Objectives

- Analyze the challenges of Business Intelligence within the company
- Compare OLTP vs. OLAP approaches
- Assess the BI tools ecosystem



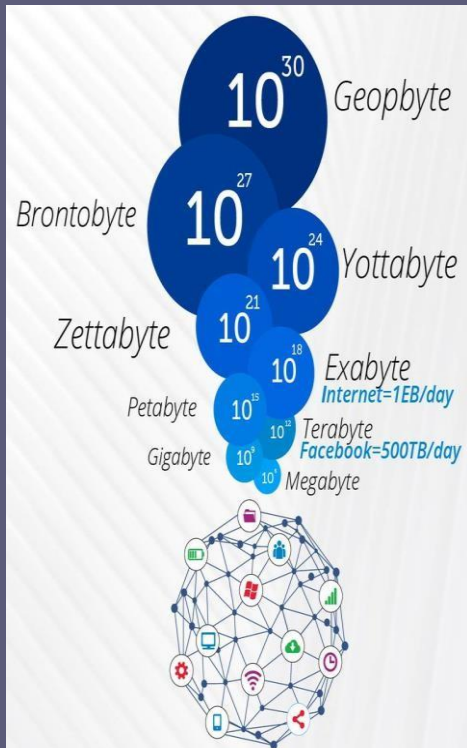
Course Plan





Introduction

Data Explosion



DATA NEVER SLEEPS 7.0

How much data is generated *every minute*?

There's no way around it: big data just keeps getting bigger. The numbers are staggering, and they're not slowing down. By 2020, there will be 40x more bytes of data than there are stars in the observable universe. In our 7th edition of Data Never Sleeps, we bring you the latest stats on how much data is being created in every digital minute.



SOURCES: STATISTA, INTERNET LIVE STATS, EXPANDED RAMBLINGS, NATIONAL ASSOCIATION OF CITY TRANSPORTATION OFFICIALS, WIRED



THE DATA EXPLOSION



Websites

Every day, 2.5 quintillion bytes of data are created (IBM, 2021)



Mobile

By 2025, the global data sphere is expected to reach 175 zettabytes (IDC)



Transactions

80 % of business data is unstructured (emails, videos, social media)



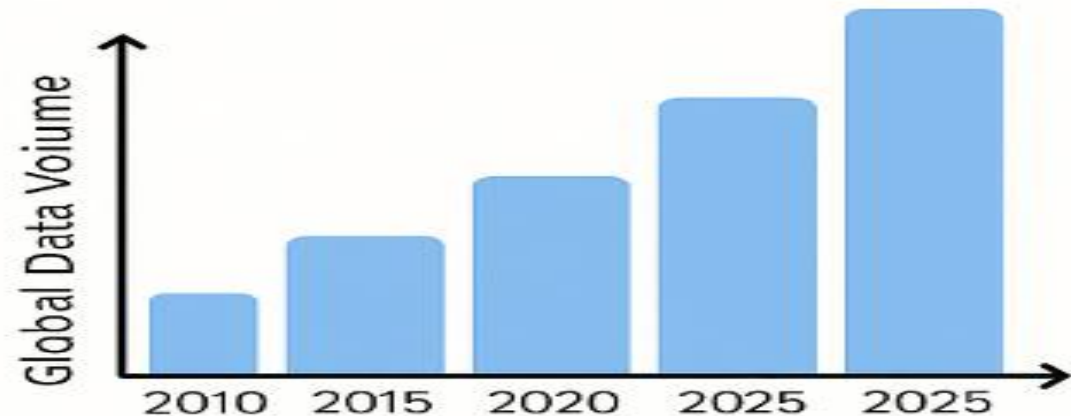
Social Media



Financial

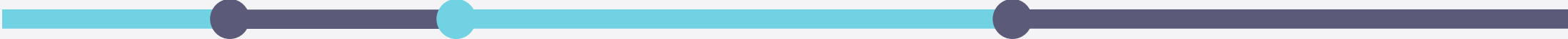


Source



How can we turn this raw data into business value?

- 
- In today's era of digital transformation, companies generate and collect massive volumes of data: sales, customer interactions, social media, internal processes, IoT sensors, and more. This explosion of data represents both an **opportunity** and a **challenge**.
 - **The challenge:** data is often scattered, heterogeneous, and difficult to exploit in its raw form. Without the right tools, decision-makers struggle to extract clear and reliable insights.
 - **The opportunity:** when properly leveraged, data enables organizations to anticipate needs, improve operational efficiency, better target customers, and foster innovation.
 - This is where **Business Intelligence (BI)** comes in. BI provides organizations with methods and solutions to:
 - transform raw data into meaningful information,
 - facilitate and accelerate decision-making,
 - optimize overall performance and increase competitiveness.
 - Thus, BI is not just a technical tool but a **strategic enabler**, driven by the need to **make sense of data** and maximize **Return on Investment (ROI)** in an increasingly complex and competitive environment.



Definition and Challenges of BI

Understand what Business Intelligence is, why it matters for organizations, and how it has evolved over time to create business value.



BI and BA: Two Complementary Analytical Approaches

- There is often confusion between Business Intelligence and Business Analytics. And it's no surprise: the two concepts are closely related and serve the same objective—simplifying the decision-making process.
- In fact, BA relies on Business Intelligence to be implemented. Every dashboard, report, or ad hoc query generated through BI contributes to Business Analytics.
- Data analysis is a core component of both BI and BA, but they use different programs and are applied for different purposes.
- While BI **delivers operational data and facts (who, what, when, how much)**, BA (sometimes called “business analytics”) **provides deeper insights that help explain and better understand those facts.**

Definition of Business Intelligence (BI)

- Business Intelligence (BI), also called **informatique décisionnelle in French**, is a set of processes, technologies, skills, and applications used to deliver relevant information to executives and managers within organizations. Its goal is to analyze large volumes of data stored in data warehouses or datamarts and transform them into actionable insights.
- Organizations that implement Business Intelligence in their decision-making process adopt a so-called **data-driven strategy**.



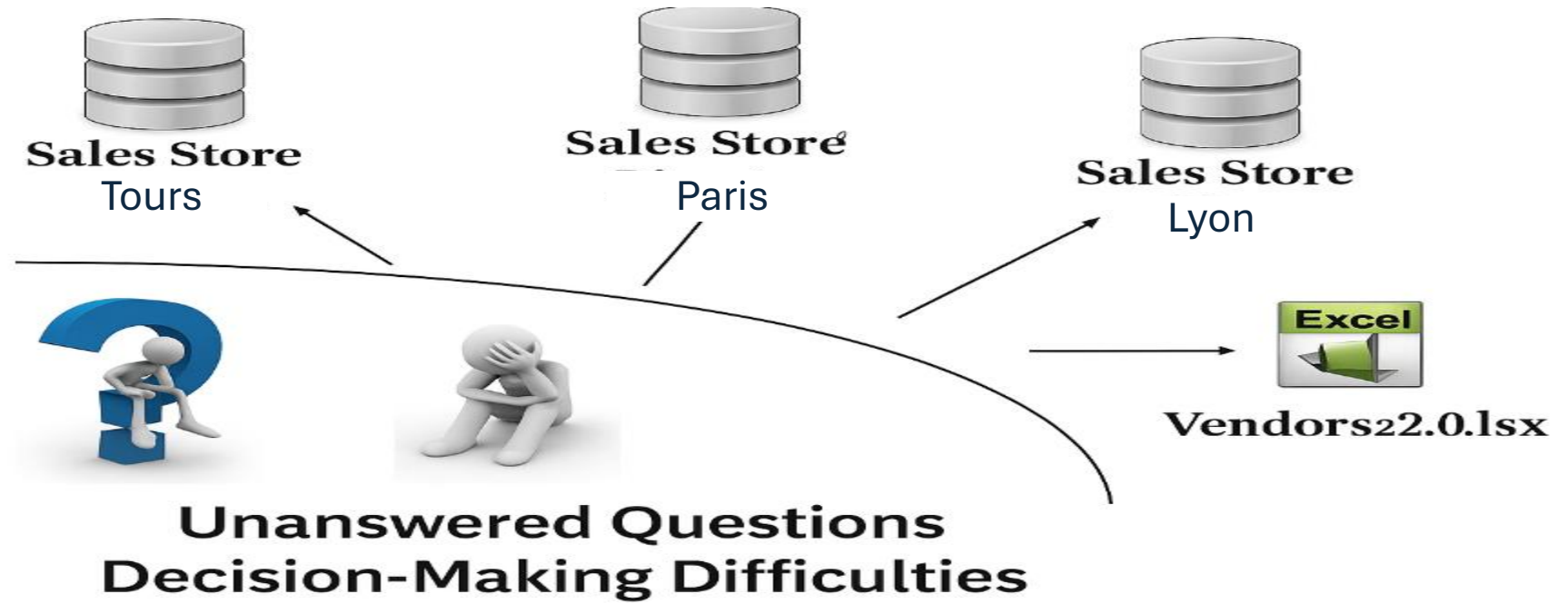
What is Business Intelligence?



Business Intelligence (BI) is a set of technological processes used to collect, manage, and analyze organizational data in order to generate insights that support a company’s strategies and operations.



Why build a BI ?



What are the SALES results:

- By product range,
- By vendor grade,
- By region,
- And by store for the second week of September of last year?

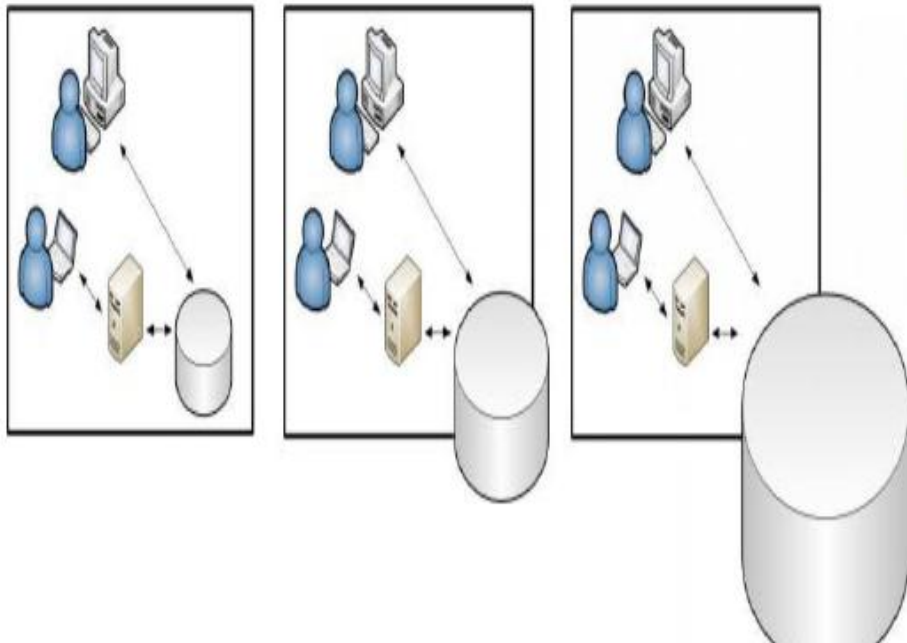


Why build a BI ?

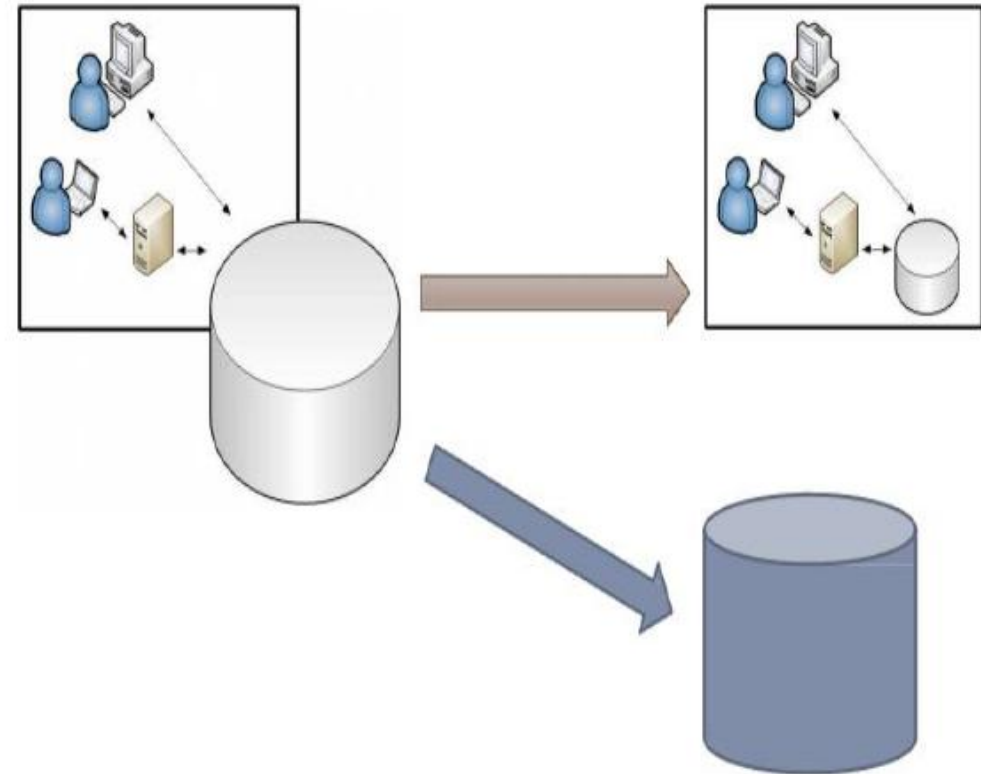
- **Production information systems do not answer the decision-makers' questions mentioned earlier because:**
 - The data sources are too complex (databases, files, etc.).
 - Queries are costly.
 - The operational system cannot be interrupted.
 - There is no data traceability.
 - The company's data is scattered.

Why build a BI ?

First Requirement

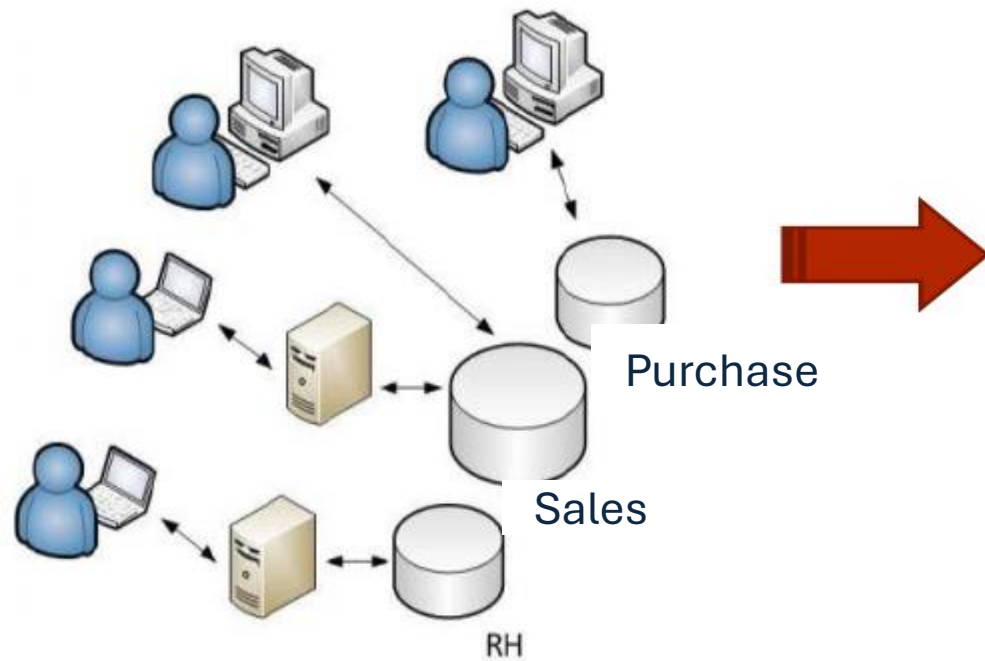


Data History (Historization)

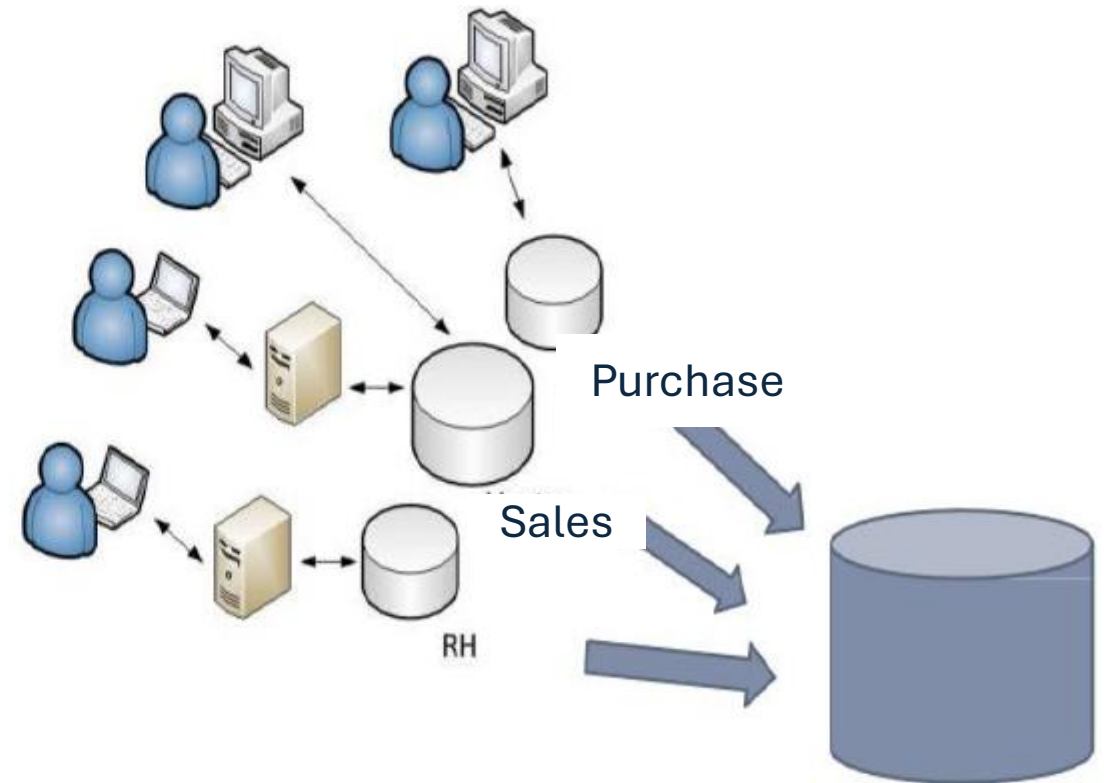


Why build a BI ?

Second Requirement

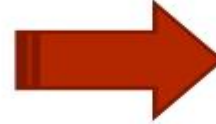
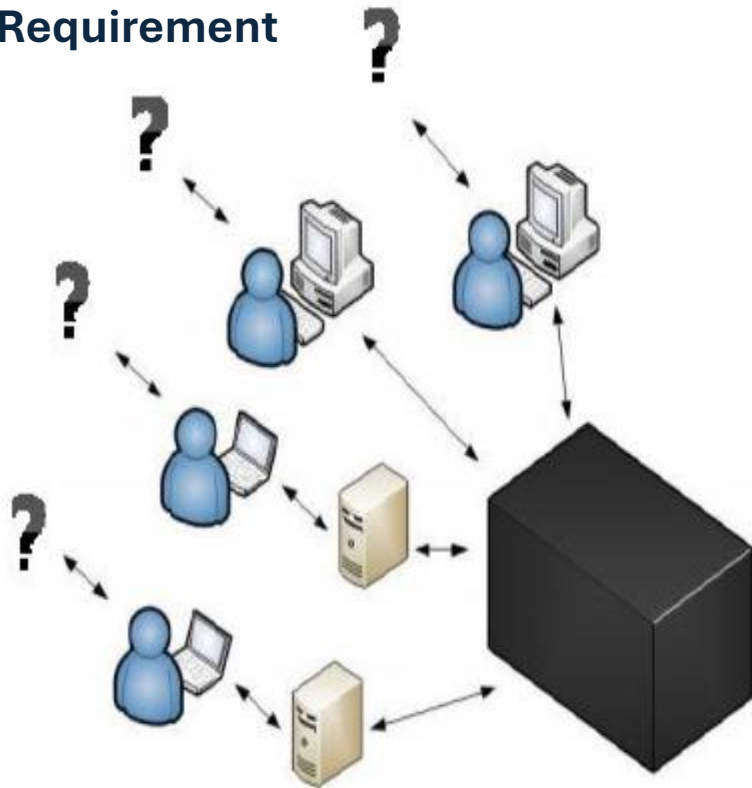


Centralization

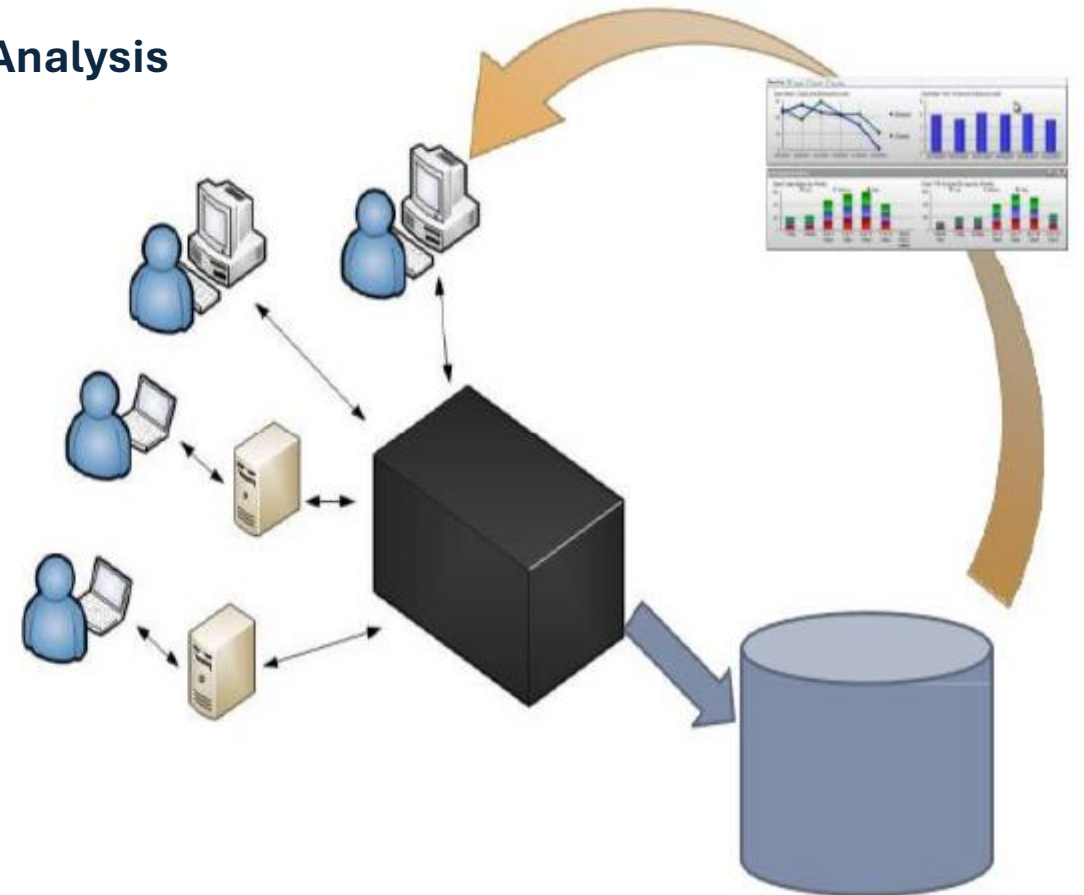


Why build a BI ?

Third Requirement



Analysis



Goals of BI

- Facilitate decision-making and guide the development of future activities.
- Deliver the right information to the right people at the right time.
- Help organizations understand their market environment.
- Generate customized reports for each department or activity.
- Support data-driven strategies through DataPrep (data preparation) and DataViz (clear and simplified visualization).



Challenges of BI

❑ Business Intelligence can be applied across **multiple domains**, including:

- **Market data:** statistics, social media, competitor analysis, etc.
- **Marketing data:** website traffic, consumer behavior, etc.
- **Financial data:** billing, cash flow, etc.
- **Human resources:** employee absences, lateness rates, etc.

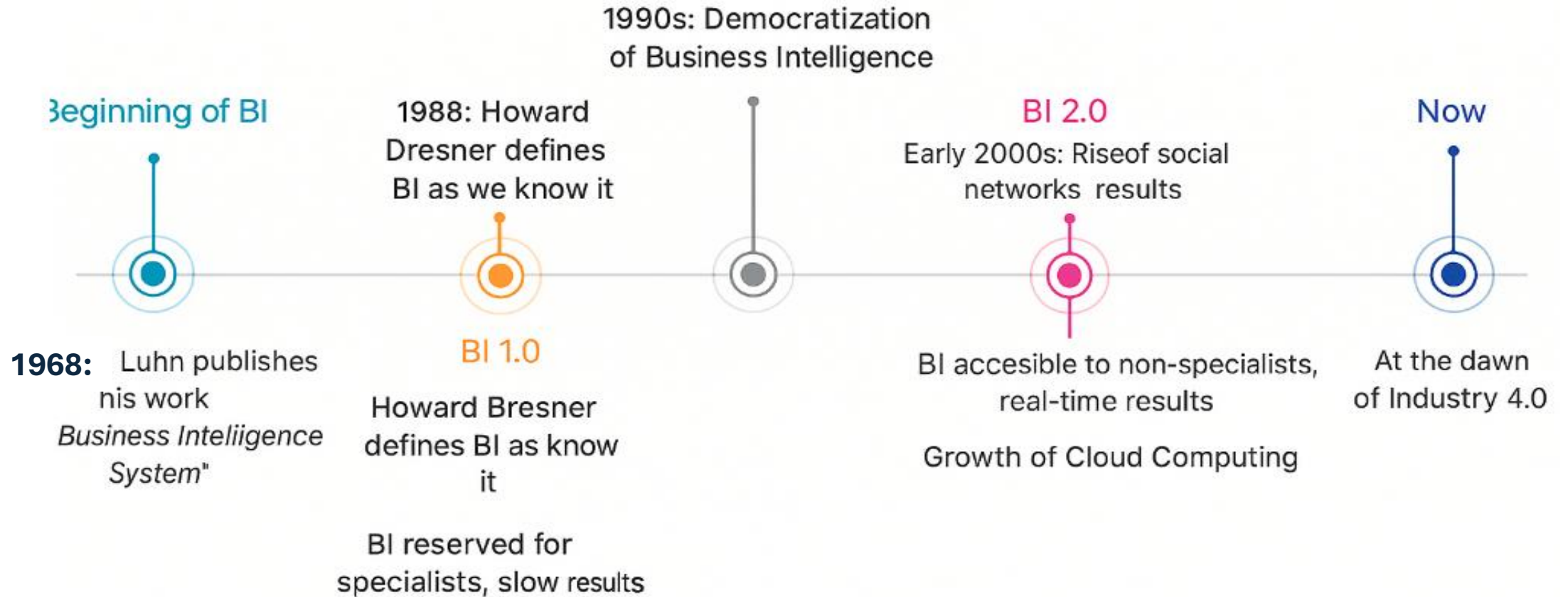




History of Business Intelligence

- **1865:** First use of the term by Richard Millar Devens, describing a banker who used market information to gain a strategic edge.
- **1958:** Hans Peter Luhn (IBM) explored automating information collection and analysis — foundations of early BI platforms.
- **1960s–1970s:** Emergence of Data Management Systems and *Decision Support Systems (DSS)* → basis for modern BI tools: OLAP, EIS, Data Warehouses.
- **1990s:** BI gained popularity in businesses but remained complex. Users relied on IT for reports, and analysts required extensive training.
- **2000s:** Rise of *Self-Service BI* → business users create their own reports. Growth of data visualization tools (Tableau, QlikView). Stronger integration with ERP/CRM.
- **2010s:** Big Data era → BI handles massive, real-time, diverse data. Cloud BI enables flexible, collaborative access. AI/ML integrated for predictive & prescriptive analytics. Popular tools: Power BI, Tableau, Qlik, Looker.
- **2020s–2025:** BI merges with Data Analytics & Data Science. Focus on augmented analytics (AI-assisted exploration), real-time BI (streaming, IoT), and data democratization. Unified platforms combine reporting, visualization, Big Data, AI, and governance.

Evolution of business intelligence



Advantages/ Disadvantages of BI

- ✓ Clear and simplified reporting with interactive dashboards.
- ✓ Data-driven decisions based on accurate business insights.
- ✓ Consolidated data from multiple internal and external sources.
- ✓ Improved efficiency by detecting and removing bottlenecks.
- ✓ Deeper insights into customers, markets, and performance.
- ✓ Faster, more informed decision-making.
- ✓ Higher customer and employee satisfaction.



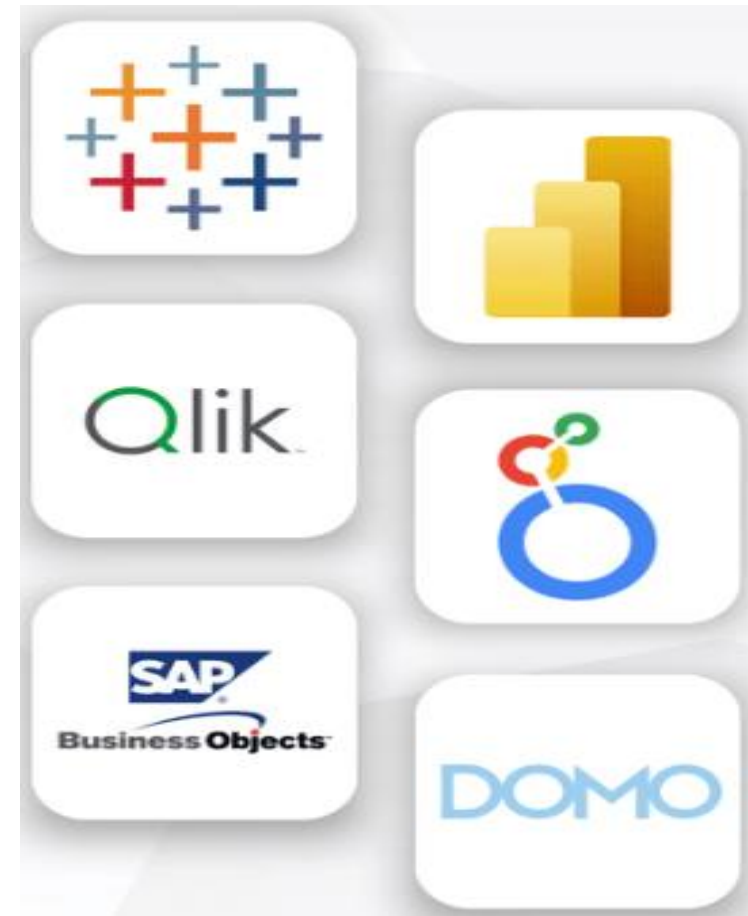
Advantages/ Disadvantages of BI

- ✓ High costs: implementation, software licenses, infrastructure, and maintenance can be expensive.
- ✓ Technical complexity: often requires specialized skills for integration, setup, and usage.
- ✓ Data dependency: incomplete, inaccurate, or poorly integrated data can lead to biased analyses.
- ✓ Implementation time: BI projects may take months or even years before delivering real value.
- ✓ Resistance to change: employees or managers may hesitate to adopt new tools and methods.
- ✓ Information overload: too much data or too many reports can complicate decision-making.
- ✓ Security risks: centralizing and sharing sensitive data increases risks of breaches and cyberattacks.



Most Popular Business Intelligence Tools

- Business Intelligence uses different tools to collect, store, analyze, and visualize business data. The most popular tools are data visualization tools, real-time BI tools, and automated reporting tools. Here are some of them:

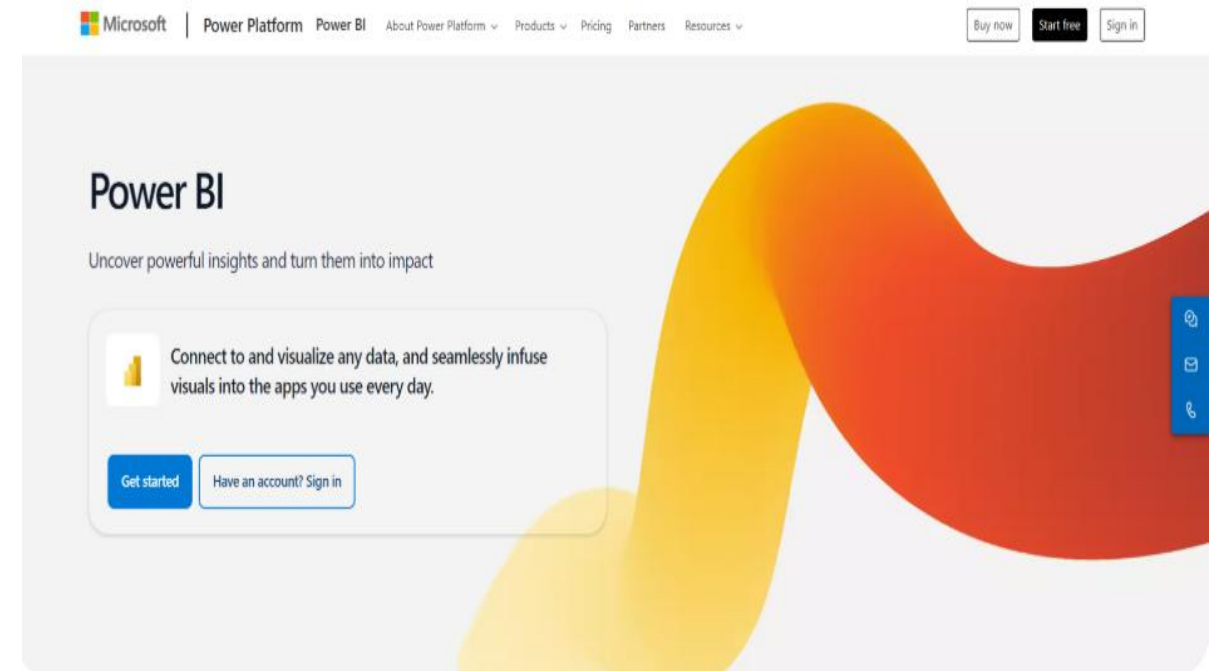


Power BI

Microsoft Power BI is another top contender in the business intelligence space. It offers robust data modeling and reporting capabilities, integrated seamlessly with other Microsoft products. Power BI's cloud-based service allows users to access and share reports from anywhere, fostering collaboration across the organization.

Key Features:

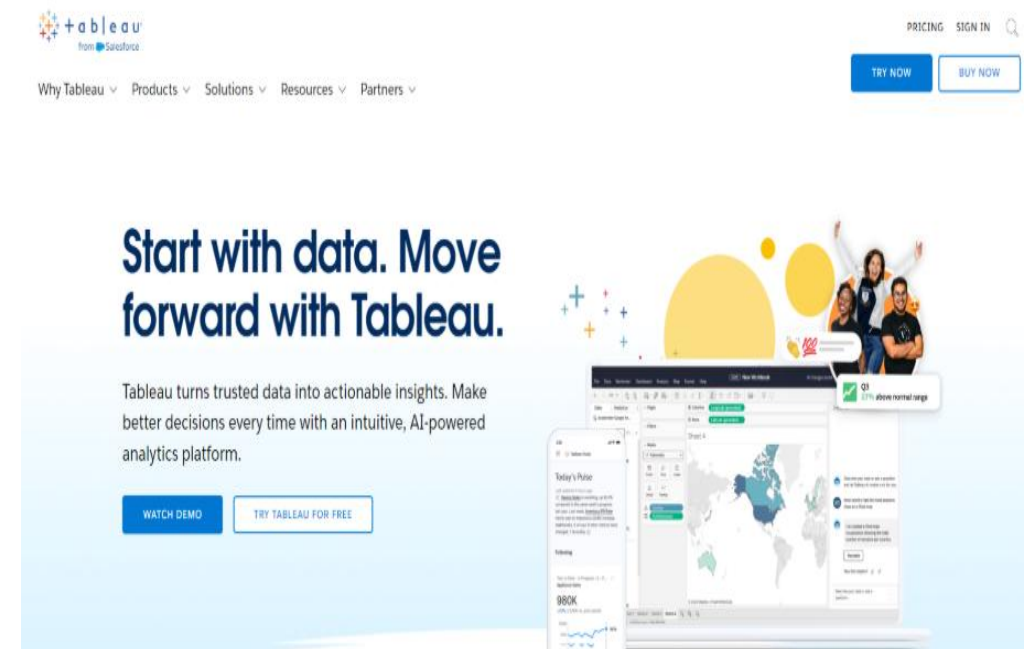
- Interactive dashboards and reports
- Seamless integration with Microsoft Office 365
- AI-powered data insights
- Extensive data connectivity



Why Choose Power BI? Power BI is ideal for organizations already using Microsoft products, providing a cohesive and integrated data analysis environment.

Tableau

- **Tableau** is a leading business intelligence tool known for its powerful data visualization capabilities. It allows users to create interactive and shareable dashboards that present data in a visually appealing manner.
- Tableau connects to various data sources, including on-premises and cloud databases, making it a versatile choice for businesses of all sizes.
- **Key Features:**
 - Drag-and-drop interface
 - Real-time data analysis
 - Integration with numerous data sources
 - Advanced analytics with AI capabilities
- **Why Choose Tableau?** Tableau's intuitive design and strong community support make it easy for users to adopt and create compelling data stories.

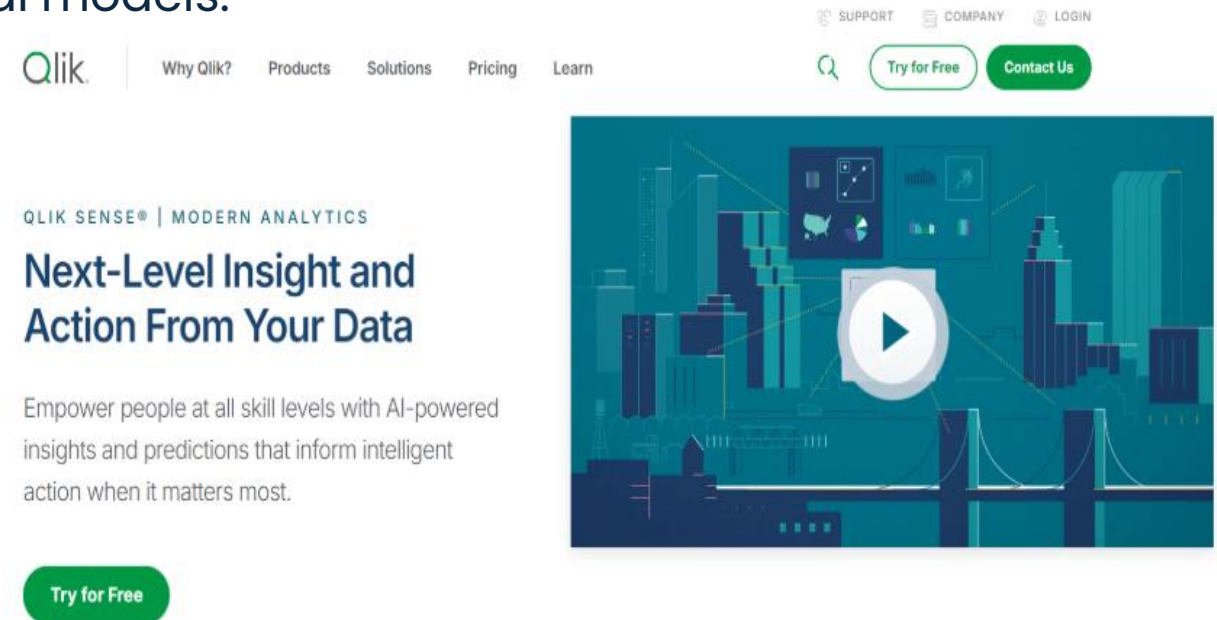


Qlik Sense

Qlik Sense is a self-service business intelligence tool that focuses on data discovery and visualization. It empowers users to explore data and uncover insights without needing extensive technical skills. Qlik's associative data model allows for more flexible and dynamic data exploration compared to traditional hierarchical models.

Key Features:

- Associative data model
- AI-powered insights and recommendations
- Easy data integration
- Customizable dashboards



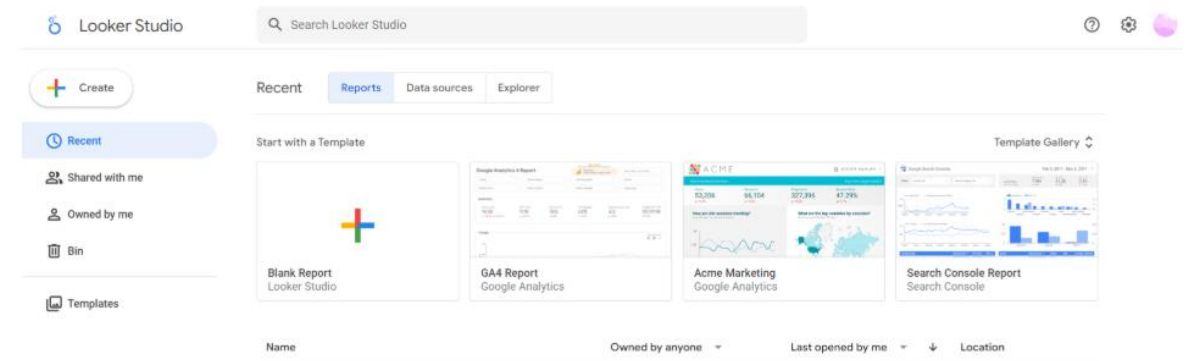
Why Choose Qlik Sense? Qlik Sense's unique approach to data exploration helps users uncover hidden patterns and insights, making it a favorite among data enthusiasts.

Looker

Looker, now part of Google Cloud, offers a modern approach to business intelligence with its web-based platform. It excels in data exploration and real-time analytics, enabling organizations to leverage their data for better decision-making. Looker's LookML language allows for customizable data models tailored to specific business needs.

Key Features:

- Real-time data exploration
- Customizable data models with LookML
- Integration with Google Cloud services
- Scalable architecture



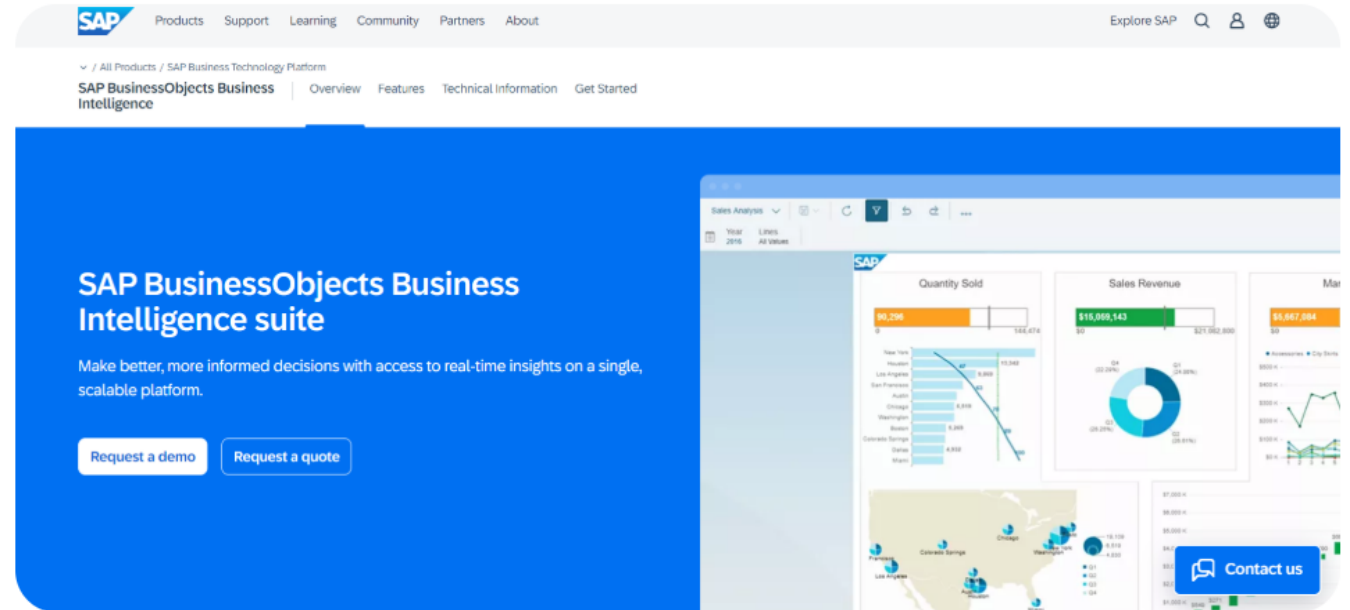
Why Choose Looker? Looker's integration with Google Cloud makes it an excellent choice for businesses looking to harness the power of cloud analytics and data warehousing.

SAP Business Objects

SAP Business Objects is a comprehensive suite of business intelligence tools designed for enterprise-level analytics. It offers a range of applications for reporting, data visualization, and in-depth analysis, all integrated within the SAP ecosystem. This tool is well-suited for large organizations with complex data needs.

Key Features:

- Enterprise-level reporting and analysis
- Integration with SAP applications
- Real-time data access
- Scalable and secure platform



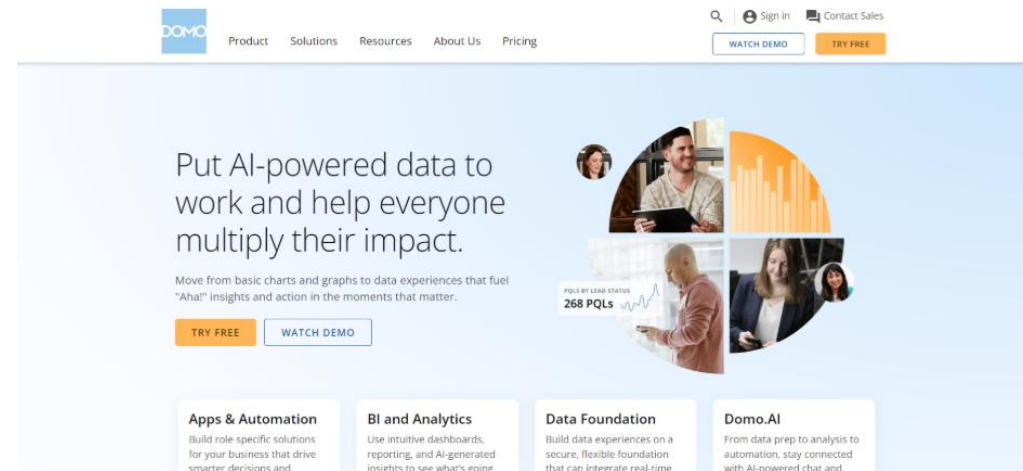
Why Choose SAP BusinessObjects? For businesses already using SAP solutions, BusinessObjects provides a seamless and powerful BI solution that integrates well with existing systems.

Domo

Domo is a cloud-based business intelligence platform that focuses on real-time data integration and collaboration. It allows users to connect to multiple data sources and create interactive dashboards with ease. Domo's mobile-friendly design ensures that users can access insights on the go.

Key Features:

- Real-time data integration
- Collaborative platform with social features
- Mobile-first design
- Extensive app marketplace



Why Choose Domo? Domo's real-time capabilities and collaborative features make it a great choice for organizations looking to foster data-driven cultures.



How BI Drives ROI

- In today's digital age, companies across every industry are overwhelmed with data — customer data, sales numbers, social media interactions, and operational metrics, just to name a few. Yet, many businesses still struggle to turn this raw data into actionable insights that can boost profitability.
- This is where Data Analytics and Business Intelligence (BI) step in — not just as reporting tools, but as strategic engines for driving Return on Investment (ROI).
- Data Analytics focuses on exploring data, identifying patterns, and predicting future outcomes.
- Business Intelligence (BI) transforms these insights into dashboards, reports, and decision-support systems that help organizations act with precision.

ROI: Return on Investment

- **ROI(Return on Investment):** is a performance metric used to evaluate the efficiency of an investment. It measures the profit or gain generated from an investment relative to its cost. A high ROI indicates the investment's gains compare favorably to its cost, making it a good measure for project justification and prioritization.

Example:

If a company invests €100,000 and earns €150,000, then:

$$ROI = \frac{\text{Gains} - \text{Costs}}{\text{Costs}} \times 100$$

$$ROI = \frac{150,000 - 100,000}{100,000} \times 100 = 50\%$$

✓ The company achieved a **50% ROI** on its investment.





How BI Influences ROI

Business Intelligence (BI) improves ROI by enabling organizations to:

- **Reduce costs** → by identifying waste and inefficiencies (e.g., optimizing inventory).
- **Increase revenue** → by better understanding customers and personalizing offers.
- **Improve decision-making speed** → with real-time dashboards and analytics.
- **Lower risks** → by relying on accurate data instead of intuition.

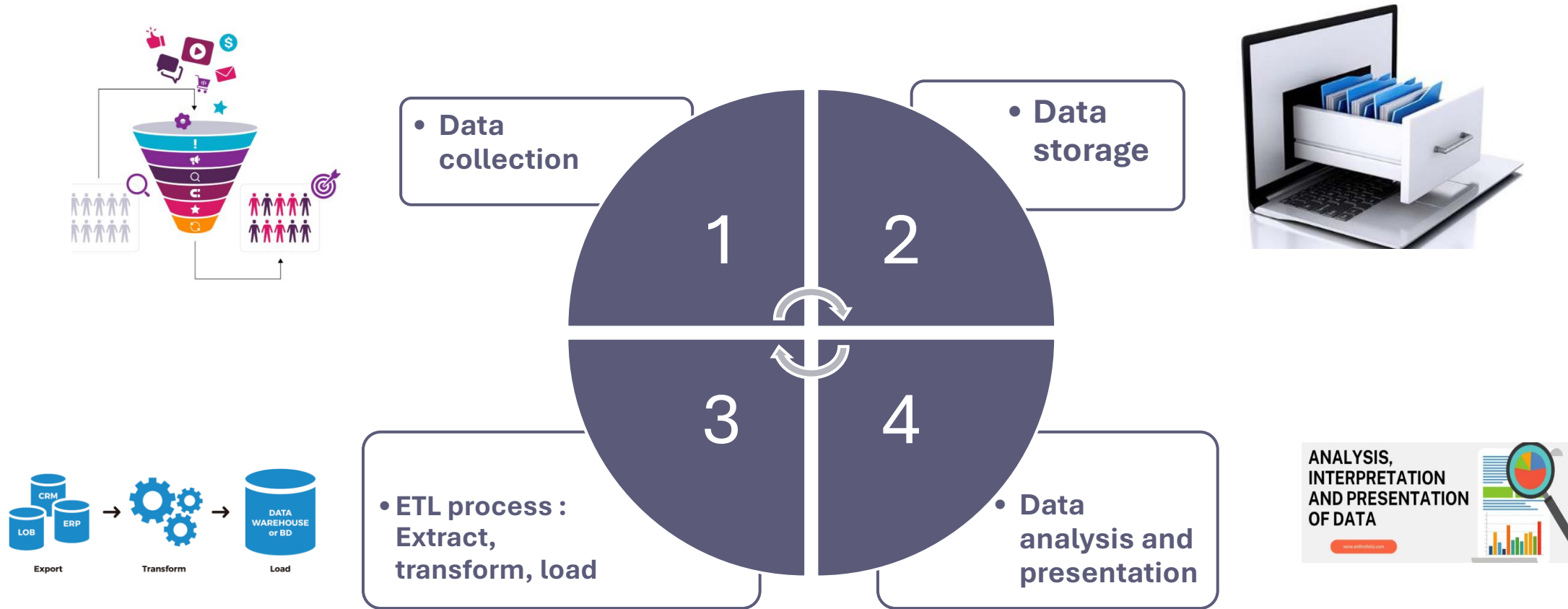
Simple Example

A supermarket using BI:

- **Without BI** → orders too many products that don't sell → losses.
- **With BI** → analyzes sales history and adapts orders → less waste, more sales → **higher ROI**.

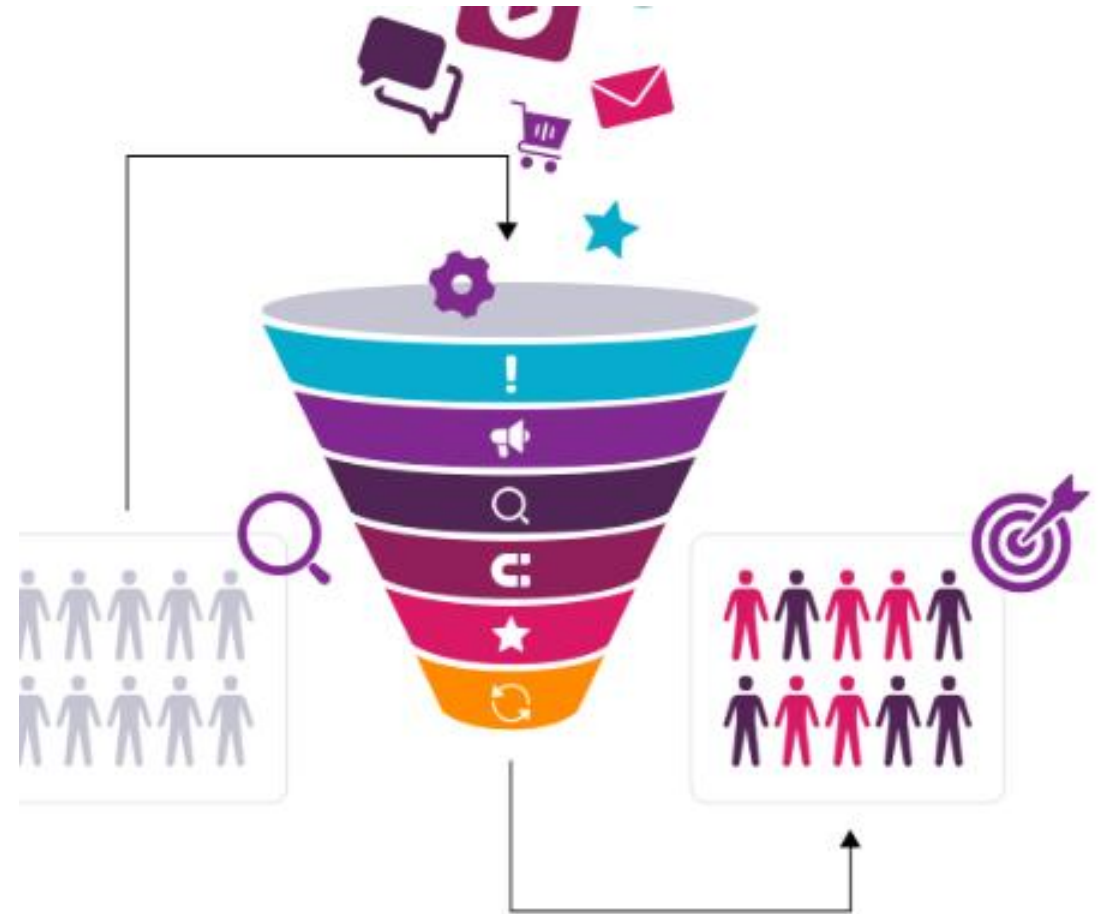
Business Intelligence (BI) Architecture

➤ A **Decision Support Information System (BI)** includes four main components:



Data collection

- Data collection is a crucial step that consists of gathering information from multiple operational sources within the company. These data are often heterogeneous and non-standardized. The main data sources may include:
 - Operational production systems:** Flat files from the operating system, database systems, ERP (Enterprise Resource Planning).
 - Archives:** Necessary for long-term analysis as they provide historical data.
 - Internal data:** Such as spreadsheets and individual workbooks, which are not directly linked to the company's operational systems.
 - External data:** Such as periodic reports from external sources, competitive analysis information, journals, etc.



ETL process : Extract, transform, load

➤ It is a system through which all the data from operational systems will pass before reaching the desired form in the warehouse. The output data will be cleaned, refined, contextualized, and ready to be stored in the warehouse.

- This is the first step, which consists of identifying data from different sources. It involves reading and understanding the data source, then extracting the required data and directing it toward the decision support system.

Extraction



After the extraction, the data undergo a transformation process that includes:

- **Cleaning:** Filtering missing data, correcting errors, removing duplicates, resolving conflicts.
- **Formatting and standardization:** Defining data types and field lengths.
- **Merging or splitting composite information.**
- **Assigning surrogate keys.**

Transformation



This is the final step, which allows the data to be loaded into the *presentation layer* of the decision support system. There are two types of loading:

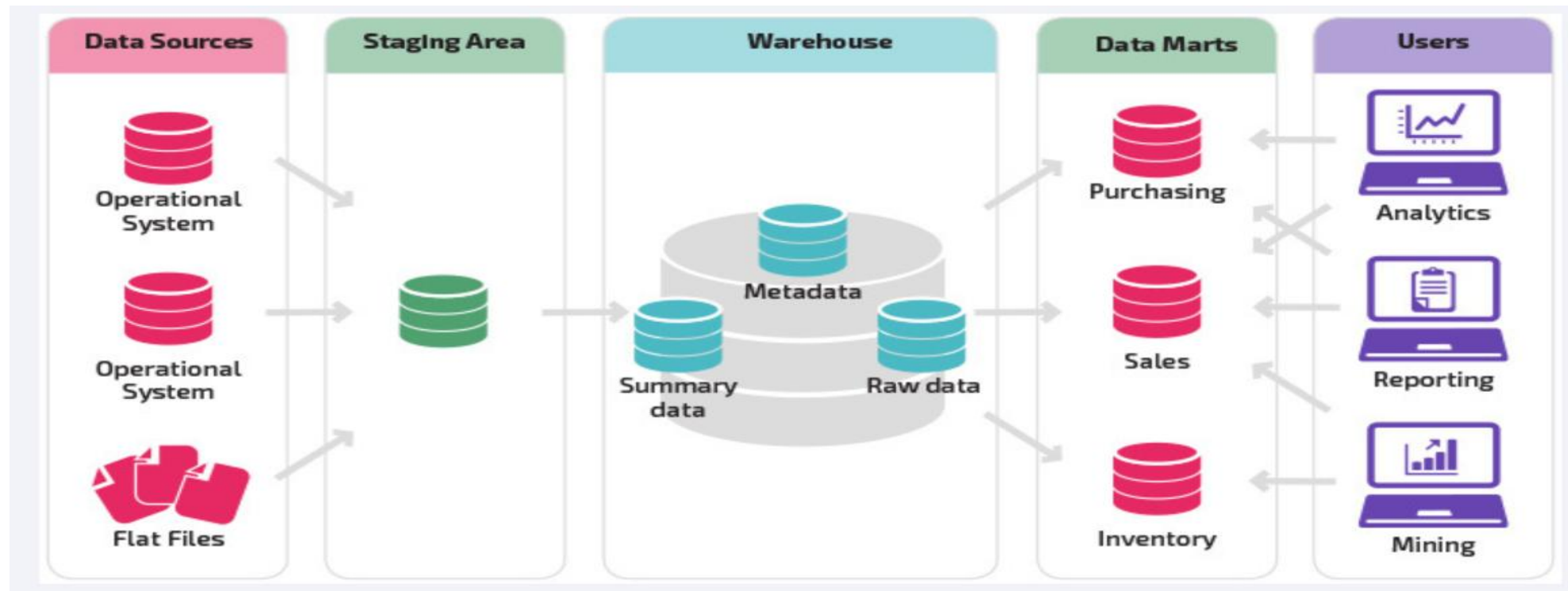
- **Initial loading:** The first load of the data warehouse, and in special cases such as data loss, where all data must be reloaded.
- **Incremental loading:** Adding new data to the existing warehouse. This operation is performed periodically.

Loading



Data Storage

- A data warehouse is a means of centralizing a large volume of data in a single unified location that is accessible to all users. Two types of data warehouses can be distinguished: Data Warehouse and Data Mart.



Data Storage

Aspect	Data Mart	Data Warehouse
Focus	A single subject or functional organization area	Enterprise-wide repository of disparate data sources
Data Sources	Relatively few sources linked to one line of business	Many external and internal sources from different areas of an organization
Size	Less than 100 GB	100 GB minimum, often terabytes in large organizations
Normalization	No preference between normalized and denormalized structures	Mostly denormalized for faster querying and performance
Decision Types	Tactical decisions (specific business lines and operations)	Strategic decisions (affecting the entire enterprise)
Cost	Typically from \$10,000 upwards	Often greater than \$100,000 (lower for cloud pay-per-use models)
Setup Time	3–6 months	At least 1 year on-premise; much quicker with cloud solutions
Data Held	Typically summarized data	Raw data, metadata, and summary data



Data analysis and presentation

To facilitate access to information for all users according to their business profiles, and to extract decision-making elements in order to enhance the overall responsiveness of the company, certain tools have been made available to decision-makers. These tools are applications that can be either simple (Reporting, Dashboards) or more sophisticated (Data Mining).



OLTP vs OLAP: Comparative Analysis

Compare transactional systems (OLTP) and analytical systems (OLAP) to understand their distinct purposes and applications

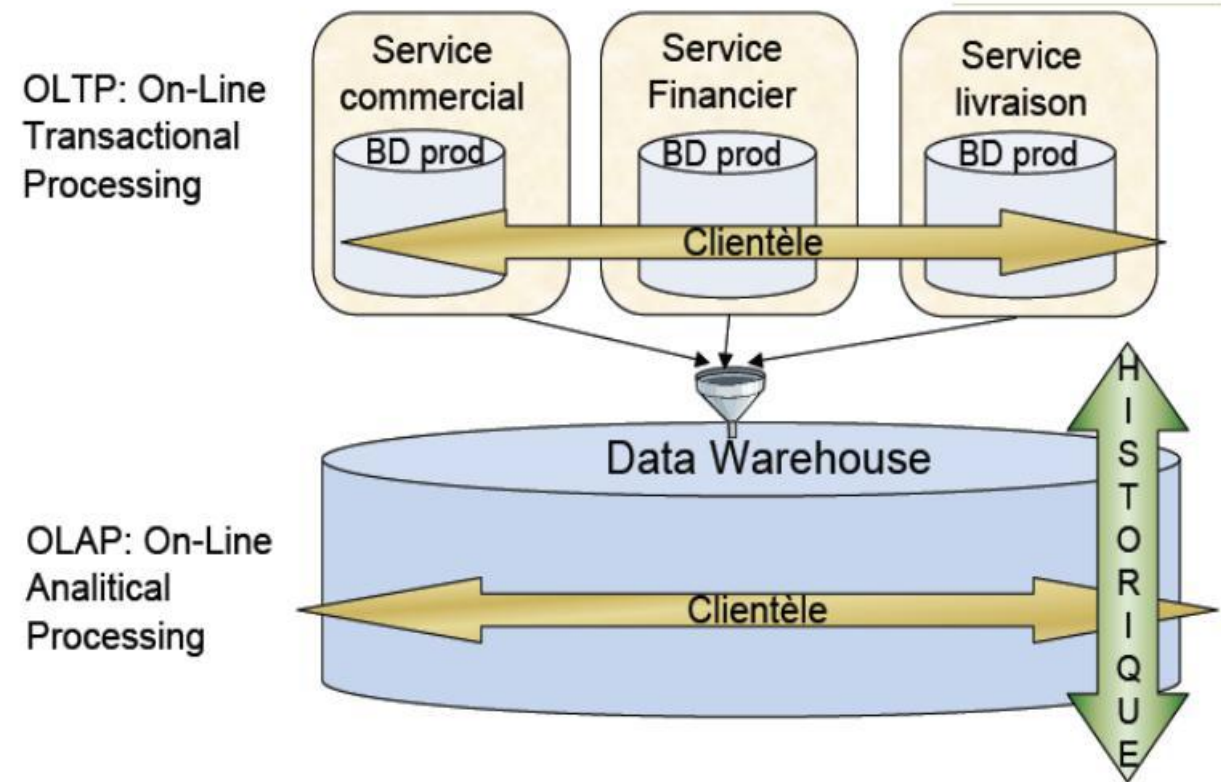
Definition of OLTP/OLAP

➤ Online Analytical Processing (OLAP) and Online Transaction Processing (OLTP) are data processing systems that help you store and analyze business data. You can collect and store data from multiple sources, such as websites, applications, smart meters, and internal systems.

➤ **OLAP** combines and aggregates data so that you can analyze it from different perspectives.

➤ **OLTP**, on the other hand, stores and updates high-volume transactional data in a reliable and efficient way.

➤ **OLTP databases can be one of the many data sources for an OLAP system.**





OLTP – Online Transaction Processing

A database management system designed to record and manage daily transactions in real-time, optimized for fast transaction processing.

Key Characteristics

- Focus on fast transaction processing.
- Used in transactional systems (banking, POS, reservations, e-commerce).
- Handles high volume of small transactions (insert, update, delete).
- Normalized data structure to reduce redundancy and ensure consistency.
- Optimized for write operations (insert/update/delete).
- Provides low latency → real-time data availability.

Examples

- Supermarket checkout system (POS).
- Online banking.
- Airline ticket reservations.
- E-commerce order management.



OLAP (Online Analytical Processing)

A database system designed for analytical purposes, optimized to handle complex queries on large volumes of data for reporting and decision-making.

Key Characteristics

- Focus on complex data queries (multi-dimensions, hierarchies).
- Supports multidimensional analysis (time, product, region, etc.).
- Used in analytical systems (BI tools, data warehouses, decision support).
- Manages a low volume of large transactions (batch updates).
- Optimized for read operations (reporting, dashboards, data mining).
- Typically has high data latency (data updated periodically, not real-time).

Examples of OLAP Systems

- Retail: Analyze sales by product, region, and season.
- Banking & Finance: Customer segmentation by income, age, and spending patterns.
- Healthcare: Analyze patient treatment outcomes by age group and illness type.
- Manufacturing: Identify bottlenecks in production across multiple plants.
- Telecommunications: Analyze call patterns and network usage by region.
- Marketing: Evaluate campaign performance by demographics and channel.



OLTP – Online Transaction Processing

- Online Analytical Processing (OLAP) and Online Transaction Processing (OLTP) are data processing systems that help you store and analyze business data. You can collect and store data from multiple sources, such as websites, applications, smart meters, and internal systems.

Main Differences: OLAP vs. OLTP

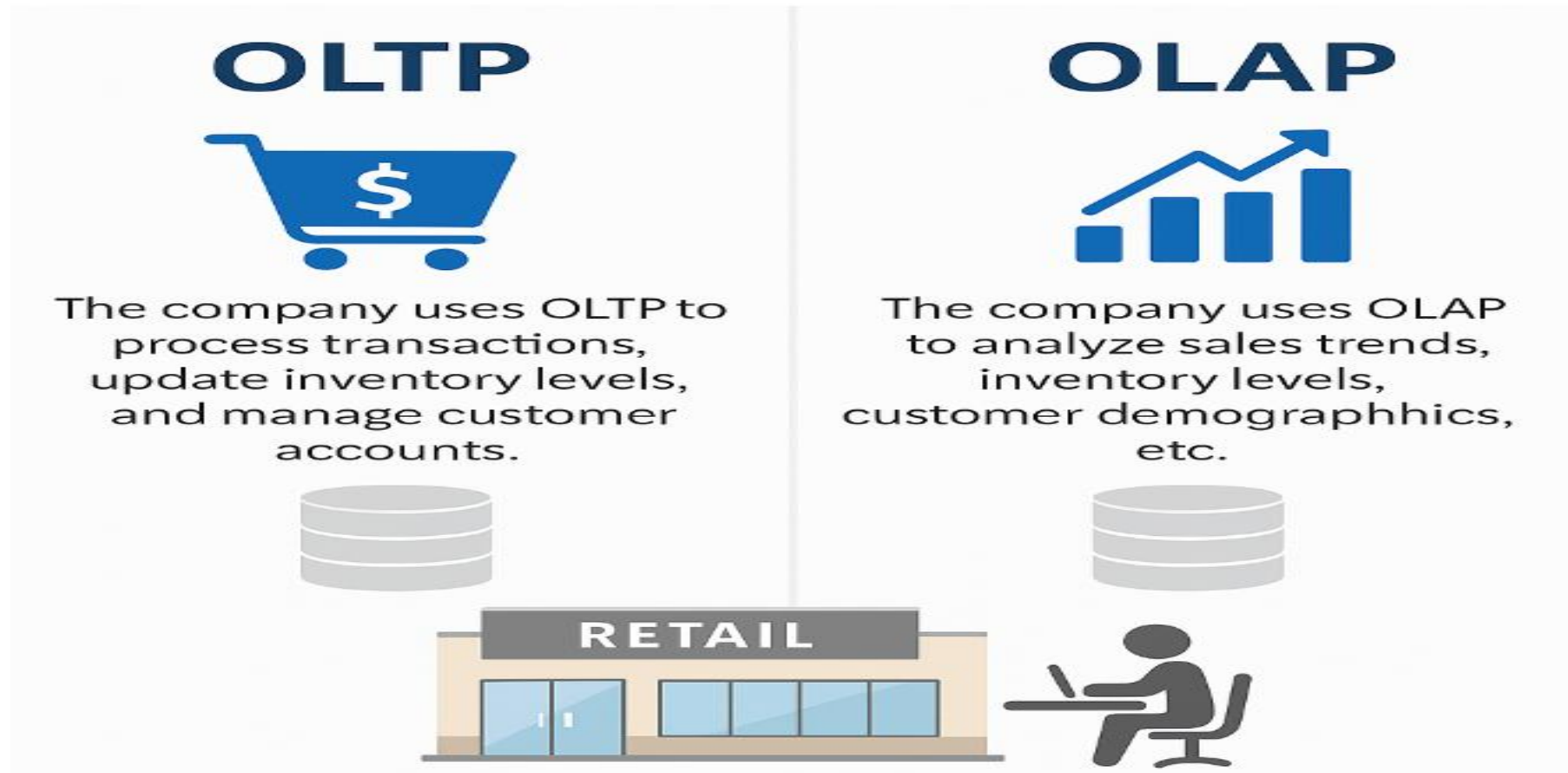
Aspect	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Main Goal	Analyze aggregated data, identify trends, support decision-making	Process day-to-day transactions reliably and efficiently
Users	Managers, analysts, decision-makers	Operational staff, clerks, end-users
Data Model	Multidimensional (cubes, dimensions, measures)	Relational (tables, rows, columns)
Focus	Read-intensive (complex queries, large volumes of historical data)	Write-intensive (frequent updates of transactional data)
Architecture	Optimized for queries and analysis, lower priority on availability	Optimized for high availability and data integrity
Performance	Query results: minutes to hours, updated in large periodic batches	Real-time processing: milliseconds, frequent short transactions
Storage Size	Terabytes to petabytes (centralized data warehouses)	Gigabytes (operational databases)
Decision Type	Strategic (long-term insights, enterprise-wide)	Tactical/Operational (daily operations, orders, payments, stock levels)



Example of Comparison Between OLAP and OLTP

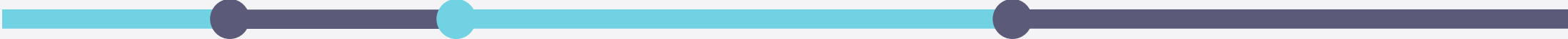
- Let's take the example of a large retail company that operates hundreds of stores across the country. The company has a massive database that tracks sales, inventory, customer data, and other key indicators.
- The company uses OLTP to process real-time transactions, update inventory levels, and manage customer accounts. Each store is connected to the central database, which updates inventory levels in real time as products are sold. The company also uses OLTP to manage customer accounts, such as tracking loyalty points, handling payment information, and processing returns.
- At the same time, the company uses OLAP to analyze the data collected by OLTP. Business analysts can use OLAP to generate reports on sales trends, inventory levels, customer demographics, and other key indicators. They run complex queries on large volumes of historical data to identify patterns and trends that inform business decisions. For example, they may identify popular products during a specific period and use this information to optimize inventory budgets.

Example of Comparison Between OLAP and OLTP



When to Use OLAP vs. OLTP

Use Case	OLAP (Online Analytical Processing)	OLTP (Online Transaction Processing)
Main Purpose	Data analysis, reporting, decision support	Real-time transaction processing
When to Use	- Generating business reports - Identifying trends - Performing complex queries - Strategic planning	- Processing sales orders - Updating inventory - Managing customer accounts - Handling payments/returns
Data Type	Historical, aggregated, multidimensional	Current, detailed, transactional
Decision Type	Strategic / Tactical	Operational / Day-to-day
Typical Users	Managers, business analysts, executives	Cashiers, clerks, customer service agents



Conclusion



Conclusion

- **Business Intelligence (BI)** provides methods and tools to transform raw data into actionable knowledge.
- BI creates value through **better decision-making, improved responsiveness, and optimized performance.**
- The **Decision-Making Architecture** relies on three layers:
 - **Operational Information Systems (OLTP)** → manage day-to-day transactions.
 - **Decision Support Systems (OLAP/SID)** → consolidate and analyze data.
 - **BI tools** → reporting, dashboards, data mining for strategic insights.
- **OLTP vs OLAP:** two complementary systems, one focused on **real-time operations**, the other on **strategic analysis**.

 **Together, they form the foundation of modern Business Intelligence.**



Questions

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